

**Web Scraping
with Python in
Azure Data
Studio**

Federal Procurement Data Systems (FPDS)

Market Research Analysis
Model



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202201-DSI-WE
March 18, 2022

Use Case - Summary

Government Contracting Market Research For Small Businesses

- Less Than 25% Of Government Contracts Are Awarded To Small Businesses
- Support Small Businesses As Engines For Economic Development
- Gain Access To The Ingenuity That Small Business Provide
- Offer Opportunities To Disadvantaged Socio-economic Groups
- Government Contracts Can Provide Residual Revenue Source Via Sole-Source Contracts

Drivers For Improvement

Major Factors Preventing Small Businesses From Competing For Federal Government Contracts

- Requirements For Contracting Are Cumbersome
- Prerequisites For Federal Contract Bidding Have Costly Overhead
- Contract Awarding Is Not Guaranteed to Have A ROI
- Market Drivers Are Vague And/Or Misleading





FFBS Solution

Fully Functional Business Solutions (FFBS)

Approach

- **Cost Effective Consultations**

- Identify the products that are the right form, fit, and function for your business
- Assist with assessments and administrative requirements for requirements gathering and generating proposals.
- Reduce costs and overhead associated the bidding for contracts.
- Remove the need for costly ranged subscriptions to public data.

Features And Benefits

- **Analytic Services**

- We remain updated on the last FAR Regulations, Contracts and Bids for every sector of business.
- In house data preparation and analytics process that will do the heavy lifting for your business.
- Our model can dually be used to bid for contracts in most states
- You own your data
- Use scalable low-cost applications to increase ROI

Marketing Analysis Dataset

Key Factors

Geographical	Contract Negotiations		Product/Service
Disasters	Price Competition	Cost Analysis	Solicitations
Population	Price Analysis	Combination Of Contract Types	Contract Awards
Crisis	Requirement Type & Complexity	Urgency	Competitors Demographics
<i>Epidemic/ Pandemics</i>	Extent and Nature of Subcontracting	Concurrency	Business Structure/Type
	Performance Period	Acquisition History	

Data Capture Part 1

Scrape Part 1 of 3

https://www.fpds.gov/ezsearch/fpdsportal?q=+SIGNED_DATE%3A%5B2022%2F03%2F05%2C2022%2F03%2F05%5D&ts=FPDS.GOV&templateName=1.5.2&indexName=awardfull&cx=30&start=0

Type one or more keywords you would like to search on:
SIGNED_DATE [2022/03/05,2022/03/05]

Contracts ICD Recovery

Result Page: 1 2 3 4 5 6 7 8 9 10 Next

Search took 0.065 seconds

You must click [here](#) for very important D&B information.

Top 10: Department Full Name

- GENERAL SERVICES ADMINISTRATION (42)
- VETERANS AFFAIRS DEPARTMENT OF (31)
- LABOR DEPARTMENT OF (18)
- STATE DEPARTMENT OF (7)
- HEALTH AND HUMAN SERVICES DEPARTMENT OF (7)
- HOME AND SECURITY DEPARTMENT OF (6)
- JUSTICE DEPARTMENT OF (5)
- INTERIOR DEPARTMENT OF THE (3)
- AGRICULTURE DEPARTMENT OF (2)
- ENVIRONMENTAL PROTECTION AGENCY (2)

Top 10: Contracting Agency Name

- FEDERAL ACQUISITION SERVICE (20)
- VETERANS AFFAIRS DEPARTMENT OF (31)
- OFFICE OF THE ASSISTANT SECRETARY FOR ADMINISTRATION AND MANAGEMENT (8)
- STATE DEPARTMENT OF (7)
- FEDERAL PRISON SYSTEM / BUREAU OF PRISONS (4)
- FOOD AND DRUG ADMINISTRATION (4)
- DEPARTMENTAL OFFICES (3)
- OFFICES, BOARDS AND DIVISIONS (2)
- ENVIRONMENTAL PROTECTION AGENCY (2)
- NATIONAL INSTITUTES OF HEALTH (2)

Top 10: Full Legal Business Name

- BRIGHT TOOL COMPANY LLC (29)
- MOND MACHINES LLC (430)
- MOORE SALES CO. INC. (41)
- USA INC. (15)
- DOCUMENT IMAGING DIMENSIONS INC. (14)
- TECHNICAL INC. (38)
- NATIONAL INDUSTRIES FOR THE BLIND (6)
- TECH CORPORATION (78)
- PREMIER & COMPANIES INC. (63)
- SAFF INC. (51)

Top 10: Treasury Account Symbol

- 4922 (6)
- 50201000 (1)
- 131430 (1)
- 145010 (1)
- 201010 (1)
- 203044 (1)
- 301000 (1)

List Of Contract Actions Matching Your Criteria

Results 1 - 30 of 2659 as of Mar 15, 2022 3:31:53 PM

Award ID (Mod#):	0003 (P00020) (View)	Award Type:	DELIVERY ORDER
Legal Business Name:	TETRA TECH INC.	Contracting Agency:	ENVIRONMENTAL PROTECTION AGENCY
Date Signed:	March 05, 2022	Action Obligation:	\$0
Referenced IDV:	EPC17046	Contracting Office:	CINCINNATI ACQUISITION DIV. (CAD)
NAICS (Code):	ADMINISTRATIVE MANAGEMENT AND GENERAL MANAGEMENT CONSULTING SERVICES (541611)	PSC (Code):	SUPPORT- PROFESSIONAL: OTHER (R499)
Entity City:	FAIRFAX	Unique Entity ID (DUNS):	180549560
Entity State:	VA	Unique Entity ID (SAM):	X38LA6SHB64
Entity ZIP:	220302201	Ultimate Parent Unique Entity ID (DUNS):	045224250
Ultimate Parent Legal Business Name:	TETRA TECH INC.	Ultimate Parent Unique Entity ID (SAM):	UMRMKLL3LGS
Cage Code:	JP0H7		
Award ID (Mod#):	1263952P0109 (0) (View)	Award Type:	PURCHASE ORDER
Legal Business Name:	COMPOSTING SME LLC	Contracting Agency:	ANIMAL AND PLANT HEALTH INSPECTION SERVICE
Date Signed:	March 05, 2022	Action Obligation:	\$38,400
Referenced IDV:		Contracting Office:	MRPES MINNEAPOLIS MN
NAICS (Code):	ANIMAL (EXCEPT POULTRY) SLAUGHTERING (311611)	PSC (Code):	ENVIRONMENTAL SYSTEMS PROTECTION- ENVIRONMENTAL REMEDIATION (F108)
Entity City:	CANDLER	Unique Entity ID (DUNS):	080869491
Entity State:	NC	Unique Entity ID (SAM):	HJ2JFCMCSHL6
Entity ZIP:	287159341	Ultimate Parent Unique Entity ID (DUNS):	080869491
Ultimate Parent Legal Business Name:	COMPOSTING SME LLC	Ultimate Parent Unique Entity ID (SAM):	HJ2JFCMCSHL6
Cage Code:	724H3		
Award ID (Mod#):	127EAS2ZF0005 (0) (View)	Award Type:	DELIVERY ORDER
Legal Business Name:	IMPERIAL FORESTRY INCORPORATED	Contracting Agency:	FOREST SERVICE
Date Signed:	March 05, 2022	Action Obligation:	\$37,352
Referenced IDV:	AG9JGPC17045	Contracting Office:	USDA-FS CSA SOUTH/WEST 2
NAICS (Code):	SUPPORT ACTIVITIES FOR FORESTRY (115310)	PSC (Code):	NATURAL RESOURCES/CONSERVATION- FOREST TREE PLANTING (F065)
Entity City:	MEDFORD	Unique Entity ID (DUNS):	809687753
Entity State:	OR	Unique Entity ID (SAM):	C8K9RS21VLG3
Entity ZIP:	975013047	Ultimate Parent Unique Entity ID (DUNS):	809687753
Ultimate Parent Legal Business Name:	IMPERIAL FORESTRY INCORPORATED	Ultimate Parent Unique Entity ID (SAM):	C8K9RS21VLG3
Cage Code:	51E00		

Search Criteria

To remove the criteria or a portion of the search criteria click the button next to each search level.

X Date Signed: [2022/03/05,2022/03/05]

Sort By

This section allows the user to sort the existing list of contracts by various fields within the contract. For example you can sort the existing list of contracts by Date Signed or Contract Type. Click on the appropriate field to Sort By. Only one Sort can be conducted at a time.

Sort Order: Descending

Relevance
Contract Type
Agency Code
Agency Full Name
Date Signed
Contracting Agency ID
Contracting Agency Name
Department Full Name
Action Obligation (\$) NAICS
PSC
Entity State
Entity ZIP Code
PoP Country Name
PoP State Name
Local Area Set Aside
Treasury Account Symbol
Non-Government Dollars
Contract Fiscal Year

Recent Searches

This section maintains a list of recent searches that were performed. Click on the desired search criteria link to conduct the search. FPDS maintains the list of searches until the clear link is selected. When you click on Clear all searches are removed.

- [2022/03/05,2022/03/05]
- [2022/01/01,2022/03/14]
- [2019/10/30,2022/02/28]
- [2022/03/05,2022/03/05] "140P3021F0168"
- PUIN="140P3021F0168" PUIN="140P3021F0168"

The screenshot displays the Azure Data Studio interface with a Python script named 'GetUrlIpynb' open in the editor. The script is designed to scrape data from the FPDS.gov website. It includes a function 'getDates' to handle date ranges and a main loop to iterate through search results. The script uses the 'requests' and 'BeautifulSoup' libraries to fetch and parse HTML data from the FPDS.gov portal. The 'EXPLORER' pane on the left shows the file structure, and the 'OUTLINE' pane at the bottom shows the script's structure. The 'TIMELINE' pane at the bottom shows the script's execution progress.

```
16
17 # getting search days from user
18 def getDates(txt):
19     msg1 = 'Enter the search ' + txt + ' in YYYY-MM-DD format'
20     startDate = input(msg1)
21     gDate = 0
22
23     while gDate != 1:
24         try:
25             Syear, Smonth, Sday = map(int, startDate.split("-"))
26             date1 = date(Syear, Smonth, Sday)
27         except:
28             startDate = input(msg1)
29             continue
30         gDate = 1
31     return date1
32 date1 = getDates('startDate')
33 date2 = getDates('endDate')
34
35 #Get new search string
36 base = date1
37 stringA = "https://www.fpds.gov/czsearch/fpdsportal?g=+SIGNED_DATE%3A%5B"
38 base = date1
39 endt = date2
40 stringB = str(base.year)+"%2F"+base.strftime("%m")+"%2F"+base.strftime("%d")+"%2C"+str(endt.year)+"%2F"+end.strftime("%m")+"%2F"+end.strftime("%d")+"%5D&s=FPDS.GOV&
41 templateName=1.5.2&indexName=awardfull&x=30&start="
42 stringC = '}'
43 myURLsearch = (stringA + stringB + stringC)
44 print(myURLsearch)
45 downloadURL = r.get(myURLsearch,headers={})
46 soup = b(downloadURL.text,"html.parser")
47 #CSS
48 fulltable_select = soup.select('span.results_heading')
49 #print(fulltable_select)
50 myLen = m.ceil(int(fulltable_select[1].find_all('b')[2].text))
51 print(myLen)
52 runInt = 1
53 wID = 0
54 while wID < myLen + 60:
55     #endt = base + timedelta(days=1)
```

Data Capture Results Part 1

DataFrame

Scrape 1 (19 Columns)

❑ **Primary Key** - N/A

❑ **Columns** -

AwardID, AwardType, LegalBusinessName, ContractingAgency, DateSigned, ActionObligation, ReferencedIDV, ContractingOffice, NAICS_Code, PSC_Code, EntityCity, UniqueEntityID_DUNS, EntityState, UniqueEntityID_SAM, Zip, UltimateParentUniqueEntityID_DUNS, UltimateParentLegalBusinessName, UltimateParentUniqueEntityID_SAM, CAGE_Code

❑ **Sample Data** - 19KE5022P0073(P00001) (View), PURCHASE

ORDER, MISCELLANEOUS FOREIGN AWARDEES, "STATE, DEPARTMENT OF", "March 05, 2022", "-\$2,577.12", , AMERICAN EMBASSY NAIROBI, HOTELS (EXCEPT CASINO HOTELS) AND MOTELS(721110), "TRANSPORTATION/TRAVEL/RELOCATION-TRAVEL/LODGING/RECRUITMENT: LODGING, HOTEL/MOTEL(V231)", WASHINGTON, 123456787, DC, LN9PU5M2YZN5, 204050001, 123456787, MISCELLANEOUS FOREIGN CONTRACTORS, LN9PU5M2YZN5, 35KC0

Data Capture Part 2

Scrape Part 2 of 3

Browser address bar: https://www.fps.gov/ezsearch/fpsportal?q=+SIGNED_DATE%3A%5B2022%2F03%2F05%2C2022%2F03%2F05%5D&s=FPDS.GOV&templateName=1.5.2&indexName=awardfull&x=30&start=30

Search bar: Type one or more keywords you would like to search on:

Navigation: [Contracts](#) [ICD](#) [Recovery](#)

Search Results Summary: Results 31 - 60 of 2659 as of Mar 15, 2022 2:18:41 PM

Buttons: [To submit comments, please click here](#)

Search Criteria:

Sort By:

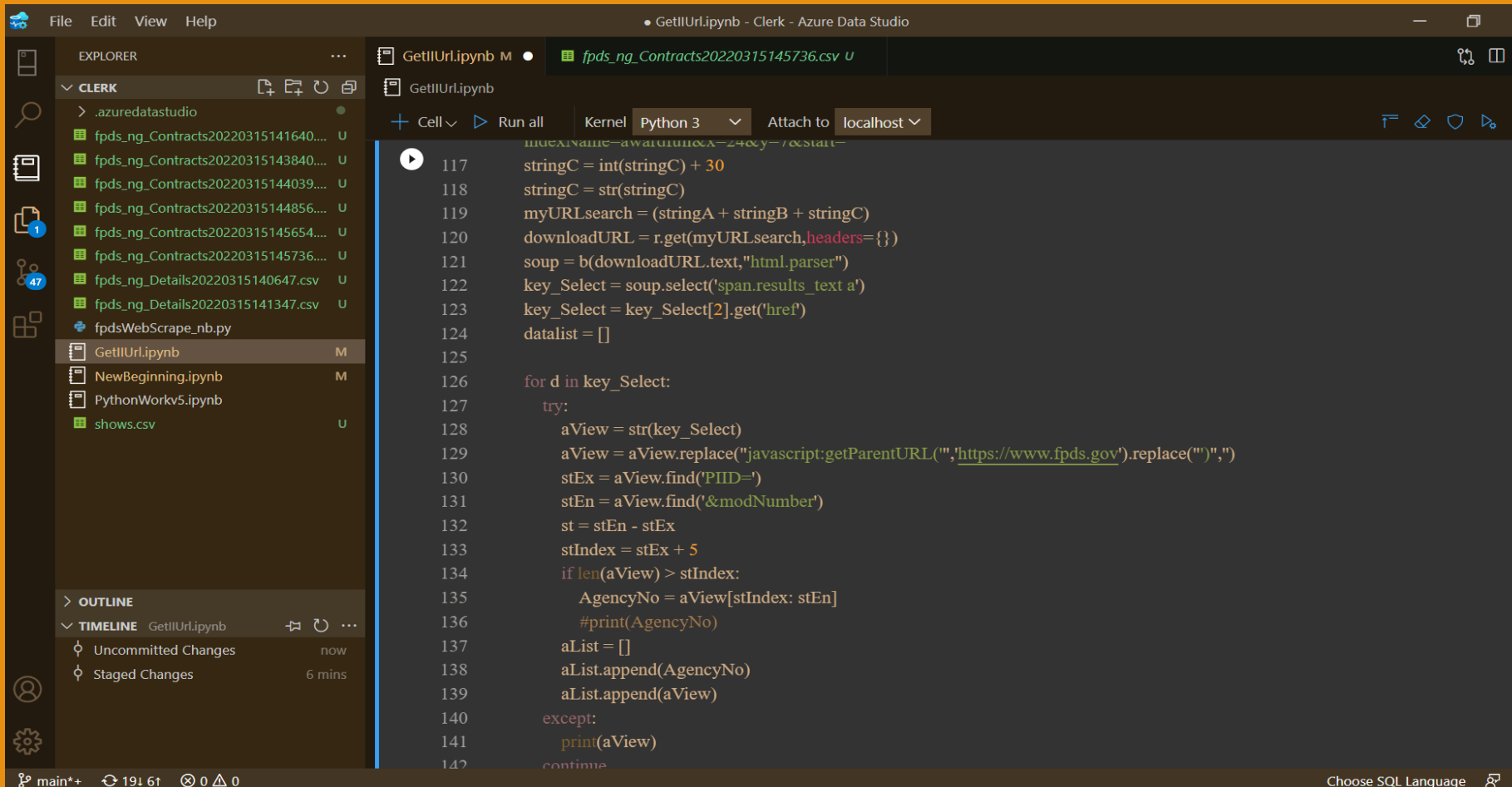
Recent Searches: [\[2022/03/05,2022/03/05\]](#) [\[2022/01/01,2022/03/14\]](#) [\[2019/10/30,2022/02/28\]](#) [\[2022/03/05,2022/03/05\] "140P3021F0168"](#) [P3021F0168 "140P3021F0168"](#)

Top 10: Department Full Name	Top 10: Contracting Agency Name	Top 10: Full Legal Business Name	Top 10: Treasury Account Symbol
- GENERAL SERVICES ADMINISTRATION (257) - VETERANS AFFAIRS DEPARTMENT OF (31) - LABOR DEPARTMENT OF (7) - STATE DEPARTMENT OF (7) - HEALTH AND HUMAN SERVICES DEPARTMENT OF (7) - HOME LAND SECURITY DEPARTMENT OF (2) - JUSTICE DEPARTMENT OF (6) - INTERIOR DEPARTMENT OF (3) - AGRICULTURE DEPARTMENT OF (2) - ENVIRONMENTAL PROTECTION AGENCY (2)	- FEDERAL ACQUISITION SERVICE (257) - VETERANS AFFAIRS DEPARTMENT OF (31) - OFFICE OF THE ASSISTANT SECRETARY FOR ADMINISTRATION AND MANAGEMENT (15) - STATE DEPARTMENT OF (7) - FEDERAL PRISON SYSTEM / BUREAU OF PRISONS (4) - FOOD AND DRUG ADMINISTRATION (4) - DEPARTMENTAL OFFICES (2) - OFFICES, BOARDS AND DIVISIONS (2) - ENVIRONMENTAL PROTECTION AGENCY (2) - NATIONAL INSTITUTES OF HEALTH (2)	- WRIGHT TOOL COMPANY, LLC (879) - MONO MACHINES LLC (430) - MOBILE SALES CO., INC. (411) - GIGA, INC. (155) - DOCUMENT IMAGING DIMENSIONS, INC. (141) - PACIFIC INC. (138) - NATIONAL INDUSTRIES FOR THE BLIND (88) - NGH CORPORATION (78) - PREMIER & COMPANIES, INC. (83) - SCAP, INC. (81)	144523 (2) 720301000 (1) 131450 (1) 144521 (1) 320130 (1) 720844 (1) 151200 (1)

List Of Contract Actions Matching Your Criteria			
Award ID (Mod#): Legal Business Name: Date Signed: Referenced IDV: NAICS (Code): Entity City: Entity State: Entity ZIP: Ultimate Parent Legal Business Name: Cage Code:	19KE5022P0073 (P00001) (View) MISCELLANEOUS FOREIGN AWARDEES March 05, 2022 HOTELS (EXCEPT CASINO HOTELS) AND MOTELS (721110) WASHINGTON DC 204050001 MISCELLANEOUS FOREIGN CONTRACTORS 35K00	Award Type: Contracting Agency: Action Obligation: Contracting Office: PSC (Code): Unique Entity ID (DUNS): Unique Entity ID (SAM): Ultimate Parent Unique Entity ID (DUNS): Ultimate Parent Unique Entity ID (SAM):	PURCHASE ORDER STATE DEPARTMENT OF -\$2,577.12 AMERICAN EMBASSY NAIROBI TRANSPORTATION/TRAVEL/RELOCATION-TRAVEL/LODGING/RECRUITMENT: LODGING, HOTEL/MOTEL (7231) 123456787 LN9PUSM2Y2N5 123456787 LN9PUSM2Y2N5
Award ID (Mod#): Legal Business Name: Date Signed: Referenced IDV: NAICS (Code): Entity City: Entity State: Entity ZIP: Ultimate Parent Legal Business Name: Cage Code:	19NL8022P0315 (0) (View) TIDEWATER, INC. March 05, 2022 ELECTRONIC SHOPPING AND MAIL-ORDER HOUSES (454110) ELKRIDGE MD 210758220 TIDEWATER, INC. 1HMG8	Award Type: Contracting Agency: Action Obligation: Contracting Office: PSC (Code): Unique Entity ID (DUNS): Unique Entity ID (SAM): Ultimate Parent Unique Entity ID (DUNS): Ultimate Parent Unique Entity ID (SAM):	PURCHASE ORDER STATE DEPARTMENT OF \$19,317.59 U.S. EMBASSY THE HAGUE ELECTRICAL CONTROL EQUIPMENT (6110) 837142470 JCF9ED1AD3H8 837142470 JCF9ED1AD3H8
IDV ID (Mod#): Legal Business Name: Date Signed: Referenced IDV: NAICS (Code): Entity City: Entity State: Entity ZIP: Ultimate Parent Legal Business Name: Cage Code:	19B01022D0002 (0) (View) MISCELLANEOUS FOREIGN AWARDEES March 05, 2022 PASSENGER CAR LEASING (532112) WASHINGTON DC 204050001 MISCELLANEOUS FOREIGN CONTRACTORS 35K00	IDV Type: Contracting Agency: Action Obligation: Contracting Office: PSC (Code): Unique Entity ID (DUNS): Unique Entity ID (SAM): Ultimate Parent Unique Entity ID (DUNS): Ultimate Parent Unique Entity ID (SAM):	IDC STATE DEPARTMENT OF \$0 U.S. EMBASSY BUCHAREST LEASE OR RENTAL OF EQUIPMENT- GROUND EFFECT VEHICLES, MOTOR VEHICLES, TRAILERS, AND CYCLES (W023) 123456787 LN9PUSM2Y2N5 123456787 LN9PUSM2Y2N5

Data Capture Part 2/3 Code

Scraping Code Part 2



```
GetUrl.ipynb - Clerk - Azure Data Studio

EXPLORER
CLERK
  .azuredatstudio
  fpds_ng_Contracts20220315141640.... U
  fpds_ng_Contracts20220315143840.... U
  fpds_ng_Contracts20220315144039.... U
  fpds_ng_Contracts20220315144856.... U
  fpds_ng_Contracts20220315145654.... U
  fpds_ng_Contracts20220315145736.... U
  fpds_ng_Details20220315140647.csv U
  fpds_ng_Details20220315141347.csv U
  fpdsWebScrape_nb.py
  GetUrl.ipynb M
  NewBeginning.ipynb M
  PythonWorkv5.ipynb
  shows.csv U

OUTLINE
TIMELINE GetUrl.ipynb
  Uncommitted Changes now
  Staged Changes 6 mins

117 indexName=awardnum&start=
118 stringC = int(stringC) + 30
119 stringC = str(stringC)
120 myURLsearch = (stringA + stringB + stringC)
121 downloadURL = r.get(myURLsearch,headers={})
122 soup = b(downloadURL.text,"html.parser")
123 key_Select = soup.select('span.results_text a')
124 key_Select = key_Select[2].get('href')
125 datalist = []
126
127 for d in key_Select:
128     try:
129         aView = str(key_Select)
130         aView = aView.replace("javascript:getParentURL('','https://www.fpds.gov').replace('','')")
131         stEx = aView.find('PIID=')
132         stEn = aView.find('&modNumber')
133         st = stEn - stEx
134         stIndex = stEx + 5
135         if len(aView) > stIndex:
136             AgencyNo = aView[stIndex: stEn]
137             #print(AgencyNo)
138             aList = []
139             aList.append(AgencyNo)
140             aList.append(aView)
141         except:
142             print(aView)
143     continue
```

Data Capture Results Part 2

DataFrame

Scrape 2 (2 Columns)

❑ **Primary Key** - N/A

❑ **Columns** - AwardID, ContractUrl

❑ **Sample Data** -

19KE5022P0073, <https://www.fpds.gov/ezsearch/jsp/viewLinkController.jsp?agencyID=1900&PIID=19KE5022P0073&modNumber=P00001&transactionNumber=0&idvAgencyID=&idvPIID=&actionSource=searchScreen&actionCode=&documentVersion=1.5&contractType=AWARD&docType=B>

Data Capture Part 3

Scrape Part 3 of 3 (Section 1 of 3)

← → ↻ https://www.fpds.gov/common/jsp/LaunchWebPage.jsp?command=execute&requestid=138711280&version=1.5

Apps Bookmarks Training New Tab Gmail IVMF Edureka Sites SAS Information Ce... myPay Web Site Home - My Visual S... FullyFBS Med/Den/Vis DC_Python - Replit New American Fun... Power BI Other bookmarks Reading list

New Save Draft Validate Approve Correct Modify Save Tmplr Delete Print
Close Out Help

Transaction Information

Award Type:	Delivery/Task Order	Prepared Date:	02/23/2022 06:33:06	Prepared User:	EDELMANN.LAWRENCE@EPAMAIL.EPA.GOV
Award Status:	Final	Last Modified Date:	03/05/2022 07:29:43	Last Modified User:	EDELMANN.LAWRENCE@EPAMAIL.EPA.GOV
Closed Status:	No	Closed Status Date:		Closed By:	
		Approved Date:	03/05/2022 07:29:43	Approved By:	EDELMANN.LAWRENCE@EPAMAIL.EPA.GOV

Document Information

	Agency	Procurement Identifier	Modification No	Trans No
Award ID:	6800	0003	P00020	0
Referenced IDV ID:	6800	EPC17046	0	
Reason For Modification:	SUPPLEMENTAL AGREEMENT FOR WORK WITHIN SCOPE			
Solicitation ID:				
	Agency Identifier	Main Account	Sub Account	Initiative
Treasury Account Symbol:				Select One

Dates

Date Signed (mm/dd/yyyy) :	03/05/2022		
Period of Performance Start Date (mm/dd/yyyy) :	03/29/2018		
Completion Date (mm/dd/yyyy) :	03/31/2023		
Est. Ultimate Completion Date (mm/dd/yyyy) :	03/31/2023		
Solicitation Date (mm/dd/yyyy) :			

Amounts

	Current	Total
Action Obligation:	\$0.00	\$4,030,980.75
Base And Exercised Options Value:	\$670,110.00	\$6,224,396.00
Base and All Options Value (Total Contract Value):	\$670,110.00	\$6,224,396.00
Fee Paid for Use of IDV:	\$0.00	

Purchaser Information

Contracting Office Agency ID:	6800	Contracting Office Agency Name:	ENVIRONMENTAL PROTECTION AGENCY
Contracting Office ID:	68HERC	Contracting Office Name:	CINCINNATI ACQUISITION DIV (CAD)
Funding Agency ID:	6800	Funding Agency Name:	ENVIRONMENTAL PROTECTION AGENCY
Funding Office ID:	68S000	Funding Office Name:	REGION 3 (FUNDING OFFICE)
Foreign Funding:	Not Applicable		

Entity Information

Data Capture Part 3

Scrape Part 3 of 3 (Section 2 of 3)

← → ↻ https://www.fpds.gov/common/jsp/LaunchWebPage.jsp?command=execute&requestid=138711280&version=1.5

Apps ★ Bookmarks 📁 Training 🌐 New Tab 📧 Gmail 📧 IVMF 📧 Edureka Sites 📧 SAS Information Ce... 🌐 myPay Web Site 📧 Home - My Visual S... 📧 FullyFBS 📧 Med/Den/Vis 📧 DC_Python - Replit 📧 New American Fun... 📧 Power BI » 📧 Other bookmarks 📖 Reading list

Entity Information

SAM Exception: [Remove Exception](#)

Unique Entity ID (DUNS): 198549560

Unique Entity ID (SAM): X38LAAB5HB64

Legal Business Name: TETRA TECH, INC.

DBAN:

CAGE Code: 1P0H7

Street: 10306 EATON PL STE 340

Street2:

City: FAIRFAX

State: VA Zip: 220302201

Country: UNITED STATES

Phone: (703) 385-6000

Fax No: (703) 385-6007

Congressional District: VIRGINIA 11

Business Category

Organization Type: CORPORATE NOT TAX EXE

State of Incorporation: DE

Country of Incorporation: USA

Business Types

- ☒ Corporate Entity, Not Tax Exempt
- ☒ Relationship With Federal Government
- ☒ All Awards
- ☒ Organization Factors
- ☒ For Profit Organization

Contract Data

Type of Contract: Cost Plus Fixed Fee

Inherently Governmental Functions: Closely Associated

Multiyear Contract: No

Major Program:

National Interest Action: None

Cost Or Pricing Data: No

Purchase Card Used As Payment Method: No

Unfinalized Action: No

Performance Based Service Acquisition: Yes - Service where PBA is used

* FY 2004 and prior; 80% or more specified as performance requirement

* FY 2005 and later; 50% or more specified as performance requirement

Emergency Acquisition: Not Applicable

Contract Financing: Select One

Cost Accounting Standards Clause: Select One

Consolidated Contract: Not Consolidated

Number Of Actions: 1

Legislative Mandates

Clinger-Cohen Act: No

Labor Standards: No

Materials, Supplies, Articles, and Equip: No

Construction Wage Rate Requirements: No

Additional Reporting: Select One or More Options

- ☐ Employment Eligibility Verification (52.222-54)
- ☐ Service Contract Inventory (FAR 4.17)
- ☐ None of the Above

Interagency Contracting Authority: Not Applicable

Other Interagency Contracting Statutory Authority: (1000 characters)

Principal Place of Performance

Principal Place Of Performance Code: State Location Country

Principal Place Of Performance Code: VA USA

Principal Place Of Performance County Name: FAIRFAX CITY

Principal Place Of Performance City Name: FAIRFAX

Congressional District Place Of Performance: VIRGINIA 11

Place Of Performance Zip Code(+4): 22030 - 2201 USPS ZIP Codes

Data Capture Part 3

Scrape Part 3 of 3 (Section 3 of 3)

← → ↻ https://www.fpd.gov/common/jsp/LaunchWebPage.jsp?command=execute&requestid=138711280&version=1.5

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Product Or Service Information

Product/Service Code:	R499	Description:	SUPPORT- PROFESSIONAL: OTHER
Principal NAICS Code:	541611	Description:	ADMINISTRATIVE MANAGEMENT AND GENERAL MANAGEM
Bundled Contract:	Not Bundled		
DOD Acquisition Program:			
Country of Product or Service Origin:	USA	UNITED STATES	
Place of Manufacture:	Not a manufactured end product		
Domestic or Foreign Entity:	U.S. Owned Business		
Recovered Materials/Sustainability:	FAR 52.223-4 and FAR 52.223-9 Included OMB Policy on Sustainable Acquisition		
InfoTech Commercial Item Category:	Select One		
Claimant Program Code:			
Sea Transportation:	Select One		
GFP Provided Under This Action:	Transaction does not use GFP		
Use Of EPA Designated Products:	Meets Requirements		
Description Of Requirement: (Limit 250 characters)	Chesapeake Bay TMDL Watershed Implementation Plan and Modeling Support		
Current:	70		

Competition Information

Extent Competed For Referenced IDV:	
Extent Competed:	Full and Open Competition
Source Selection Process:	Select One
Solicitation Procedures:	Subject to Multiple Award Fair Opportunity
IDV Type Of Set Aside:	No set aside used.
Type Of Set Aside:	Select One
Type Of Set Aside Source:	IDC
Evaluated Preference:	No Preference used
SBIR/STTR:	Select One
Fair Opportunity/Limited Sources:	Fair Opportunity given
Other Than Full And Open Competition:	Select One
Local Area Set Aside:	No
Contract Opportunities Notice:	Yes
A76 Action:	No
Commercial Item Acquisition Procedures:	Commercial Item Procedures not used
IDV Number of Offers:	4
Number Of Offers Received:	3
Small Business Competitiveness Demonstration Program:	
Simplified Procedures for Certain Commercial Items:	No

Preference Programs / Other Data

Contracting Officer's Business Size Selection:	Other than Small Business
Subcontract Plan:	Individual Subcontract Plan
Price Evaluation Percent Difference:	%

Data Capture Part 3

Web Form Scrape

Transaction Information			
Award Type:	Delivery/Task Order	Prepared Date:	02/23/2022 06:33:06
Award Status:	Final	Last Modified Date:	03/05/2022 07:29:43
Closed Status:	No	Closed Status Date:	
Prepared User:	EDELIMANN.LAWRENCE@EPAMAIL.EPA.GC		
Last Modified User:	EDELIMANN.LAWRENCE@EPAMAIL.EPA.GC		
Closed By:			
Approved By:	EDELIMANN.LAWRENCE@EPAMAIL.EPA.GC		

Document Information			
Award ID:	6800	Procurement Identifier	0003
Referenced IDV ID:	6800	EPC-17046	0
Reason For Modification:	SUPPLEMENTAL AGREEMENT FOR WORK WITHIN SCOPE		
Solicitation ID:			
Agency Main Sub Identifier Account	Initiative		
Treasury Account Symbol:	Select One		

Dates			
Date Signed (mm/dd/yyyy):	03/05/2022	Action Obligation:	Current
Period of Performance Start Date (mm/dd/yyyy):	03/29/2018	Base And Exercised Options Value:	\$0.00
Completion Date (mm/dd/yyyy):	03/31/2023	Base And All Options Value (Total Contract Value):	\$670,110.00
Est. Ultimate Completion Date (mm/dd/yyyy):	03/31/2023		\$670,110.00
Solicitation Date (mm/dd/yyyy):		Fee Paid For Use of IDV:	\$0.00

Purchaser Information			
Contracting Office Agency ID:	6800	Contracting Office Agency Name:	ENVIRONMENTAL PROTECTION AGENCY
Contracting Office ID:	68HERC	Contracting Office Name:	CINCINNATI ACQUISITION DIV (CAD)
Funding Agency ID:	6800	Funding Agency Name:	ENVIRONMENTAL PROTECTION AGENCY
Funding Office ID:	68S000	Funding Office Name:	REGION 3 (FUNDING OFFICE)
Foreign Funding:	Not Applicable		

Entity Information			
SAM Exception:			
Unique Entity ID (DUNS):	198549560	Street:	10306 EATON PL STE 340
Unique Entity ID (SAM):	X38LAABSHB84	Street2:	
Legal Business Name:	TETRA TECH, INC.	City:	FAIRFAX
DBAN:		State:	VA
CAGE Code:	1P0H7	Zip:	220302201
		Country:	UNITED STATES
		Phone:	(703) 385-6000
		Fax No:	(703) 385-6007
		Congressional District:	VIRGINIA 11

Business Category			
Organization Type:	CORPORATE NOT TAX EX	Business Types	✓ Corporate Entity, Not Tax Exempt
State of Incorporation:	DE	Relationship With Federal Government	✓ All Awards
Country of Incorporation:	USA	Organization Factors	✓ For Profit Organization

Contract Data			
Type of Contract:	Inherently Governmental Functions:		
Multiyear Contract:	No		
Major Program:	None		
National Interest Action:	No		
Cost Or Pricing Data:	No		
Purchase Card Used As Payment Method:	No		
Underinitiated Action:	Yes - Service where PBA is used		
Performance Based Service Acquisition:	Not Applicable		
* FY 2004 and prior; 80% or more specified as performance requirement			
* FY 2005 and later; 90% or more specified as performance requirement			
Emergency Acquisition:	Select One		
Contract Financing:	Select One		
Cost Accounting Standards Clause:	Not Consolidated		
Consolidated Contract:	1		
Number Of Actions:			

Legislative Mandates			
Clinger-Cohen Act:	No	Principal Place Of Performance Code:	VA
Labor Standards:	No	Principal Place Of Performance City Name:	FAIRFAX CITY
Materials, Supplies, Articles, and Equip:	No	Principal Place Of Performance County Name:	FAIRFAX
Construction Wage Rate Requirements:	Select One or More Options	Congressional District	VIRGINIA 11
Additional Reporting:	Employment Eligibility Verification (S2 223-54)	Place Of Performance Zip Code(+4):	22030 - 2201
	Service Contract Inventory (FAR 4.17)		
	None of the Above		
Interagency Contracting Authority:	Not Applicable		
Other Interagency Contracting Statutory Authority:	(1000 characters)		

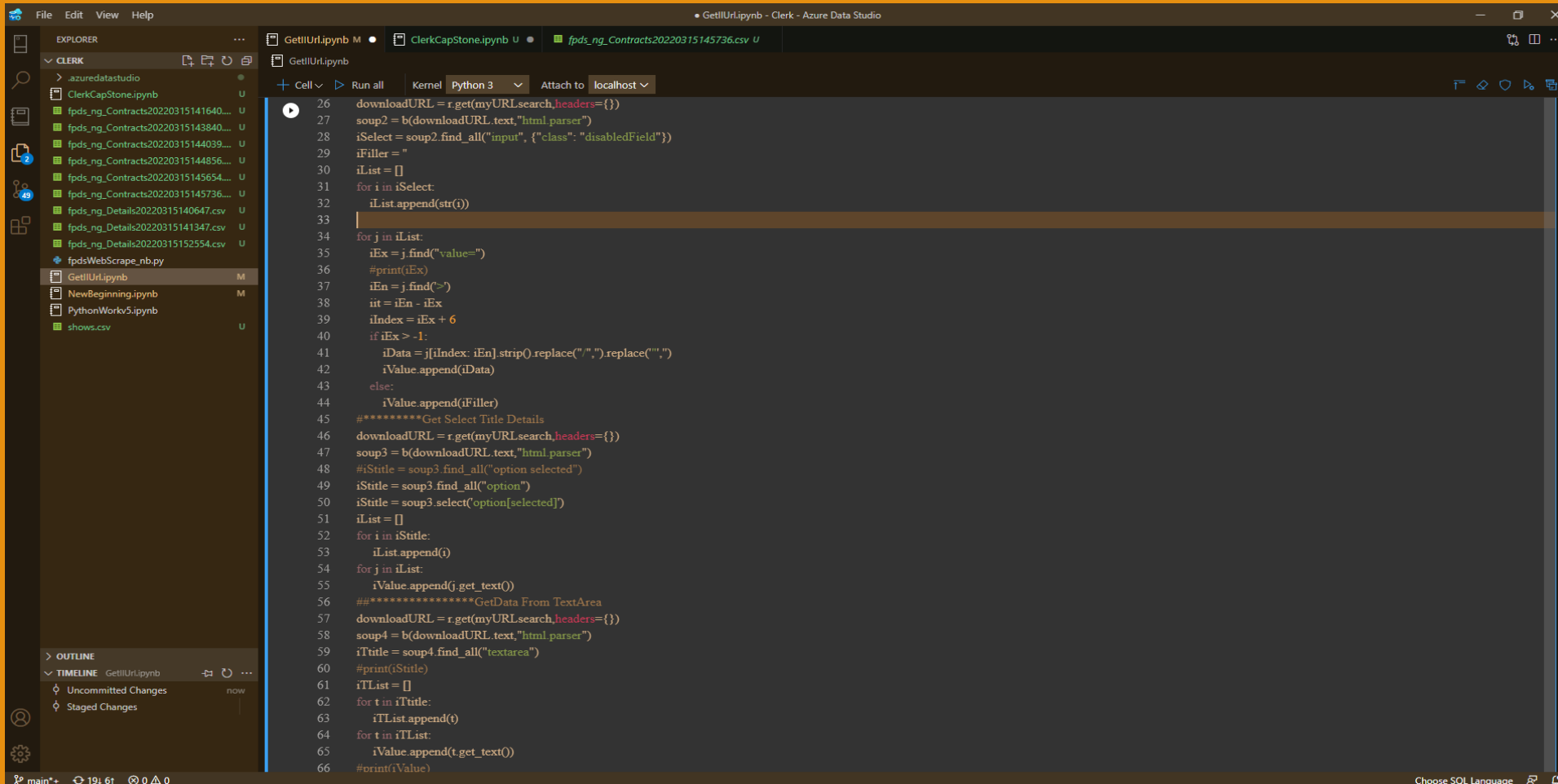
Product Or Service Information			
Product/Service Code:	R499	Description:	SUPPORT- PROFESSIONAL- OTHER
Principal NAICS Code:	541611	Description:	ADMINISTRATIVE MANAGEMENT AND GENERAL MANAG
Bundled Contract:	Not Bundled		
DOD Acquisition Program:		Description:	
Country of Product or Service Origin:	USA		
Place of Manufacture:	Not a manufactured end product		
Domestic or Foreign Entity:	U.S. Owned Business		
Recovered Materials/Sustainability:	FAR S2 223-4 and FAR S2 223-9 Included	OMB Policy on Sustainable Acquisition	
InfoTech Commercial Item Category:	Select One		
Claimant Program Code:		Description:	
Sea Transportation:	Select One		
GFP Provided Under This Action:	Transaction does not use GFP		
Use Of EPA Designated Products:	Meets Requirements		
Description Of Requirement: (Limit 250 characters)	Chesapeake Bay TMDL Watershed Implementation Plan and Modeling Support		
Current:	70		

Competition Information			
Extent Competed For Referenced IDV:			
Extent Competed:	Full and Open Competition		
Source Selection Process:	Select One		
Solicitation Procedures:	Subject to Multiple Award Fair Opportunity		
IDV Type Of Set Aside:	No set aside used		
Type Of Set Aside:	Select One		
Type Of Set Aside Source:	IDC		
Evaluated Preference:	No Preference used		
SBIR/STTR:	Select One		
Fair Opportunity/Limited Sources:	Fair Opportunity given		
Other Than Full And Open Competition:	Select One		
Local Area Set Aside:	No		
Contract Opportunities Notice:	Yes		
A76 Action:	No		
Commercial Item Acquisition Procedures:	Commercial Item Procedures not used		
IDV Number of Offers:	4		
Number Of Offers Received:	3		
Small Business Competitiveness Demonstration Program:	Number of Offers Source: This Action		
Simplified Procedures for Certain Commercial Items:	No		

Preference Programs / Other Data			
Contracting Officer's Business Size Selection:	Other than Small Business		
Subcontract Plan:	Individual Subcontract Plan		
Price Evaluation Percent Difference:	%		

Data Capture Part 3/3 Code

Scraping Code Part 3



```
26 downloadURL = r.get(myURLsearch,headers={})
27 soup2 = b(downloadURL.text,"html.parser")
28 iSelect = soup2.find_all("input", {"class": "disabledField"})
29 iFiller = "
30 iList = []
31 for i in iSelect:
32     iList.append(str(i))
33
34 for j in iList:
35     iEx = j.find("value=")
36     #print(iEx)
37     iEn = j.find(">")
38     iit = iEn - iEx
39     iIndex = iEx + 6
40     if iEx > -1:
41         iData = j[iIndex: iEn].strip().replace(",","").replace("'",")
42         iValue.append(iData)
43     else:
44         iValue.append(iFiller)
45 #*****Get Select Title Details
46 downloadURL = r.get(myURLsearch,headers={})
47 soup3 = b(downloadURL.text,"html.parser")
48 #iStitle = soup3.find_all("option selected")
49 iStitle = soup3.find_all("option")
50 iStitle = soup3.select('option[selected]')
51 iList = []
52 for i in iStitle:
53     iList.append(i)
54 for j in iList:
55     iValue.append(j.get_text())
56 ##*****Get Data From TextArea
57 downloadURL = r.get(myURLsearch,headers={})
58 soup4 = b(downloadURL.text,"html.parser")
59 iTitle = soup4.find_all("textarea")
60 #print(iStitle)
61 iTList = []
62 for t in iTTitle:
63     iTList.append(t)
64 for t in iTList:
65     iValue.append(t.get_text())
66 #print(iValue)
```

Data Capture Results Part 3

DataFrame

Scrape 3 (142 Columns)

❑ **Primary Key** - AwardID

❑ **Columns** - 'AwardType', 'PreparedDate', 'PreparedUser', 'AwardStatus', 'LastModifiedDate', 'LastModifiedUser', 'ClosedStatus', 'ClosedStatusDate', 'ClosedBy', 'Filler1', 'ApprovedDate', 'ApprovedBy', 'AwardIDAgency', 'AwardIDProcurementIdentifier', 'AwardIDModificationNo', 'TransNo', 'RefIDVIDAgency', 'RefIDVProcurementIdentifier', 'RefIDVModificationNo', 'ReasonForModification', 'SolicitationID', 'TASAgencyIdentifier', 'TASMainAccount', 'TASSubAccount', 'DateSigned'...

❑ **Row Sample** -Delivery/Task Order,05/16/2020

16:05:36,MARCIA.LEE.HE1254@HQ.DODEA.EDU,Final,05/16/2020

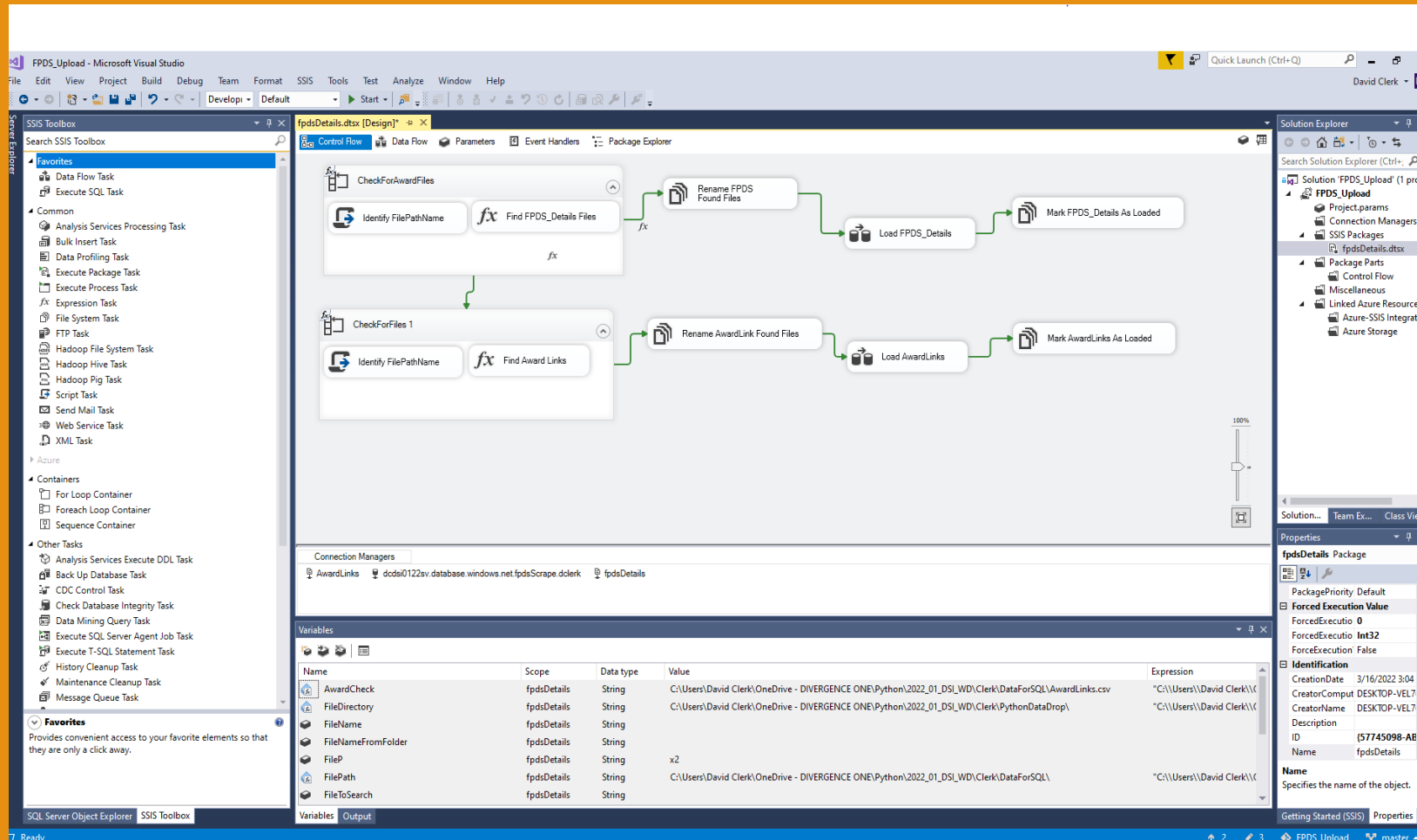
16:05:46,MARCIA.LEE.HE1254@HQ.DODEA.EDU,Yes,04/14/2020

20:21:21,FPDS_CLOSEOUT,,05/16/2020

16:05:46,MARCIA.LEE.HE1254@HQ.DODEA.EDU,9700,0061,1,0,9700,HE125415D0005,0,Funding Only Action,,97,0100,,05162020,09142016,11132016,11132016,,,-\$0.24,"\$19,999.76",,-\$0.24,"\$19,999.76",,-\$0.24,"\$19,999.76",,-\$0.24,"\$19,999.76",...

Data Wrangling Automation

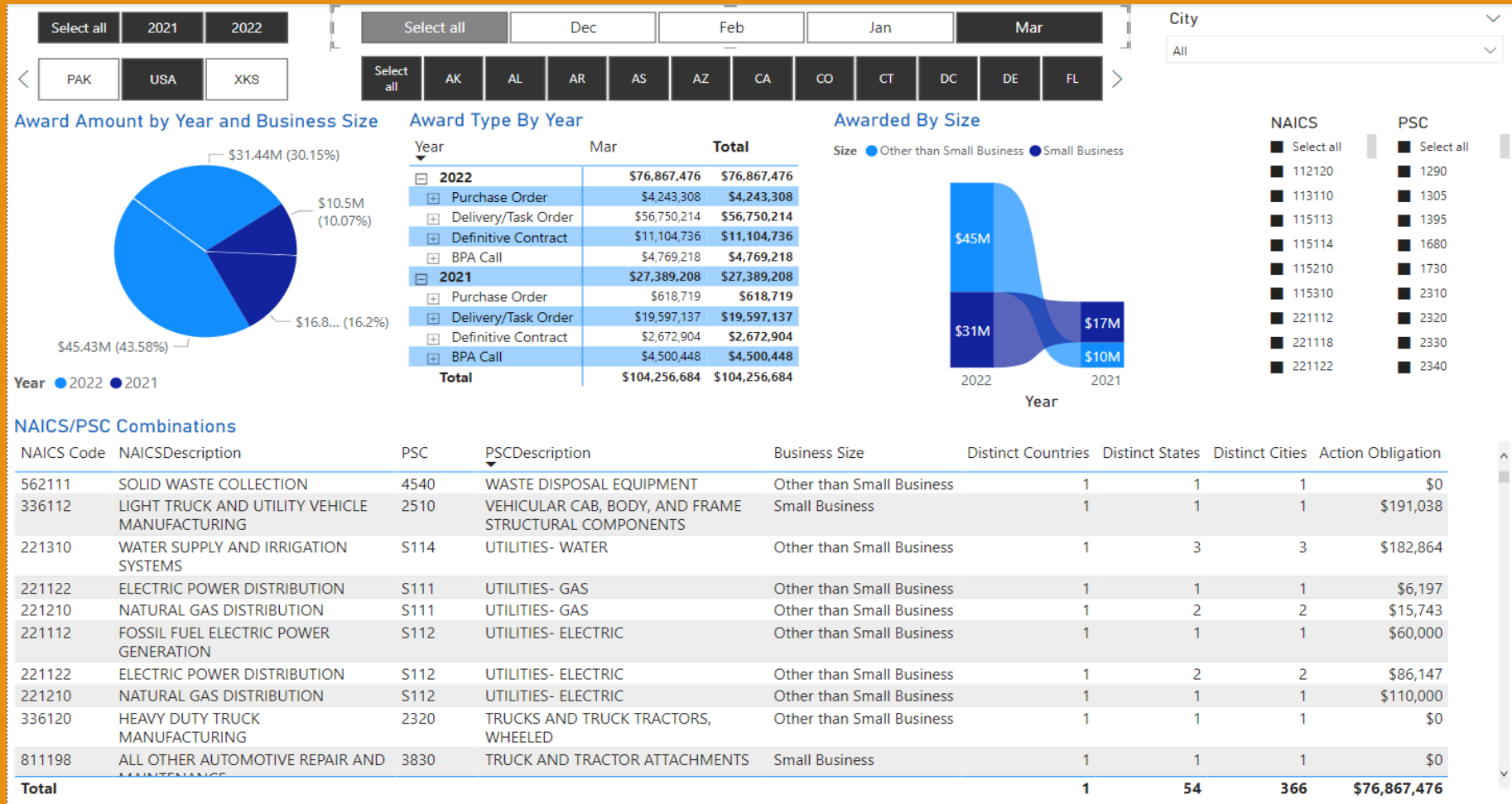
SQL Server Data Tool



After 11 Passes
thru Beautiful Soup

Reporting (Transforming Data To Information)

PowerBi Visualizations



Summary

Proof Of Concept

- **Proven From This Model**

- Obtain detailed data for public website
- Manipulate and store data on cloud/local servers.
- Automate the ETL/ELT process.
- Visualize the data to answer business queries.
- Maximize the use of coding (Python BeautifulSoup) to build models for scraping various sites/pages
- Provide analytical data for marketing, pricing strategy, etc...



Questions?

Thank You.